

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

3.1 Exploration

3.1.1 Module 3 | A Concrete Example: Cell Phones

Reflect and discuss (A) Cell Phone 1

 1. Did dialing the phone *always* result in _____? (choose one)

- A. Yes
- B. No

 2. Which of the choices best represents the _____ in the simulation? (choose one)

- A. {Phone Button [send,end]}
- B. {Phone Button [send,end], Attempts[0,10]}
- C. {Phone Button [send,end], Call Connected [yes, no]}
- D. None of the above.

Reflect and discuss (B) Cell Phones 2

Consider the following response structure:

Causal Assignment	Exercises Regularly	Smokes	Impotent = yes
1	Yes	Yes	8%
2	Yes	No	2%
3	No	Yes	10%
4	No	No	2%

 3. This response structure is _____ because: (choose one)

- A. It doesn't contain enough rows
- B. Exercise and Smoking have no influence on Impotence
- C. There are no test pairs for Exercise or Smoking
- D. There is at least one causal assignment that does not determine the value of the effect.

 4. The difference between a _____ is that: (choose one)

- A. There are no direct causes in an indeterministic response structure, while there are direct causes in deterministic response structures.

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- B. In an indeterministic response structure, there is at least one causal assignment that does not determine the value of the effect.
- C. Indeterministic response structures always have fewer variables than deterministic ones.
- D. Deterministic response structures are based on evidence, while indeterministic response are matters of speculation.

3.1.2 Module 3 | An Abstract Example: Red and Blue Squares

Reflect and discuss (C) Red Squares 1

 5. Does _____ fully determine the state of the blue square?
(choose one)

- A. Yes
- B. No
- C. Neither

 6. Does _____ fully determine the state of the blue square?
(choose one)

- A. Yes
- B. No

3.1.3 Module 3 | Individuals and Populations

Reflect and discuss (D) Individuals and Populations 1

 7. If, in a given population, we assume that every individual's response structure is deterministic, can we tell if _____? (choose one)

- A. Yes, always
- B. No, never
- C. Yes, but only in certain circumstances

Reflect and discuss (E) Individuals and Populations 2

 8. Does the causal assignment that you give to the red square have the same effect on _____?
_____ ? (choose one)

- A. Yes
- B. No

 9.

- A. The response structure for the blue square is deterministic.
- B. The response structure for the blue square is indeterministic.
- C. None of the above.

3.1.4 Module 3 | Cell Phones Revisited

Reflect and discuss (F) Cell Phones Revisited 1

 10. In which of the following causal assignments results does _____? (choose one)

- A. Situation 1: Call placed, and within tower range
- B. Situation 2: Call placed, and not within tower range
- C. Situation 3: Call not placed, and within tower range
- D. Situation 4: Call not placed, and not within tower range

 11. Is the response structure for “the call gets through” _____? (choose one)

- A. Yes
- B. No

3.1.5 Module 3 | Red and Blue Squares Revisited

Reflect and discuss (G) Red and Blue Squares Revisited 1

 12. Should the missing cell in the Response Structure for The System of Squares contain _____? (choose one)

- A. Solid
- B. Hatched

3.1.6 Module 3 | Pseudo-Indeterminism

Reflect and discuss (H) Psuedo-Indeterminism 1

Consider the response structure for malaria given in the table titled: Response Structure For Malaria – With Gene Unobserved. Suppose that we are told only that Joel was bitten by the mosquito and had no inoculation.

 13.

(choose one)

- A. Yes
- B. No
- C. Impossible to tell

Suppose that we are told that Joel was bitten by the mosquito but had an inoculation.

 14.

(choose all that apply)

- A. Assignment 1.
- B. Assignment 2.
- C. Assignment 3.
- D. Assignment 4.

- E. Assignment 5.
- F. Assignment 6.
- G. Assignment 7.
- H. Assignment 8.

If Joel was bitten by the mosquito but had an inoculation,  15.

- A. Yes
- B. No
- C. Impossible to tell

3.2 Indeterministic Causation

3.2.1 Module 3 | Cell Phones and Indeterminism

Reflect and discuss (I) Cell Phones and Indeterminism 1

 16. Suppose that, instead of flipping a computer's version of a fair coin, we programmed the simulation to roll a computer's version of a regular six sided die - and when it rolled a "1" or a "2" we set IN TOWER RANGE to equal Yes, and when it rolled either a 3,4,5, or 6 we set IN TOWER RANGE to equal No. What should we put into the box that now contains _____ in table: Response Structure for the Small Cell Phone System? (choose one)

- A. Yes = 50%, No = 50%
- B. Yes = 33%, No = 67%
- C. Yes = 67%, No = 33%
- D. Yes = 16.67%, No = 83.33%

3.2.2 Module 3 | Quantifying Indeterministic Causation

Reflect and discuss (J) Quantifying Indeterministic Causation 1

 17. Suppose there were three different media reports about an experiment on the effect of TV on violence. All the newspapers reported that the researchers gave two different causal assignments to 200 seventh grade boys. For 100 of the boys, they allowed them to watch as much violent TV as they wanted over 3 months, and for the other 100, they allowed none. The effect they measured was whether the boys got into any fights during the 3 months of the experiment. The newspapers gave different descriptions of the results. Which of the following quotes provides a fully quantified _____ for the population the researchers analyzed? (choose all that apply)

- A. "Researchers reported that the boys who were allowed to watch TV were twice as likely to get into a fight as those who were not".
- B. "Researchers reported that 12% of the boys who were allowed to watch TV got into a fight, but only 6% of the boys who were not allowed to watch got into one".
- C. "Researchers reported that the boys who were allowed to watch TV were significantly more likely to get into a fight than those who were not".
- D. None of the above.

3.3 Defining Causation with Indeterminism

3.3.1 Module 3 | Direct Causation with Indeterminism

 Reflect and discuss (K) Direct Causation with Indeterminism 1

Causal Assignment	Gender	Experience	Hired
1	Female	Yes	20%
2	Female	No	10%
3	Male	Yes	20%
4	Male	No	10%

 18. In this response structure for Hired, which are the _____ of Hired? (choose one)

- A. Gender
- B. Experience
- C. Neither Gender nor Experience
- D. Both Gender and Experience

Causal Assignment	Flu Shot	Avg. Sleep Per Night	Avoided Flu
1	Yes	Over 8 hours	90%
2	Yes	Under 8 hours	50%
3	No	Over 8 hours	20%
4	No	Under 8 hours	10%

 19. What test pairs are sufficient to determine that _____ is a direct cause of avoiding the flu in this system? (choose all that apply)

- A. 1 and 2.
- B. 1 and 4.
- C. 1 and 3.
- D. 2 and 4.

Causal Assignment	Drinks Red Wine	Drinks Coffee	Eats Red Meat	Heart Disease
1	Yes	Yes	Yes	50%
2	Yes	Yes	No	60%
3	Yes	No	Yes	50%
4	Yes	No	No	60%
5	No	Yes	Yes	70%
6	No	Yes	No	50%
7	No	No	Yes	70%
8	No	No	No	50%

20. Which of the following are direct indeterministic causes of _____, according to this indeterministic response structure? (choose all that apply)

- A. Drinks red wine.
- B. Drinks coffee.
- C. Eats red meat.
- D. None of the above.

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

21. In this case, what are the test pairs for _____? (choose one)

- A. 1 and 4, 2 and 5, 3 and 6
- B. 1 and 2, 2 and 3
- C. 1 and 2, 2 and 3, 1 and 3
- D. 4 and 5, 5 and 6, 4 and 6

Causal Assignment	Recitation Attendance	Studying Time	Got an A
1	Occasionally	High	10%
2	Moderately	High	42%
3	Frequently	High	10%
4	Occasionally	Low	10%
5	Moderately	Low	42%
6	Frequently	Low	10%

22. In this response structure, which are the _____? (choose all that apply)

- A. Recitation Attendance.
- B. Study Time.
- C. Neither Recitation nor Study Time.

3.3.2 Module 3 | Direct vs. Indirect Causation

Reflect and discuss (L) Direct vs. Indirect Causation 1

Consider the following response structure for the set of variables {Speeds Excessively [yes, no], Traffic Infraction [yes, no], Lose License [yes, no]}, where Lose License is treated as the effect.:

Causal Assignment	Speeds Excessively	Traffic Infraction	Lose License = yes
1	Yes	Yes	65%
2	Yes	No	1%
3	No	Yes	65%
4	No	No	1%

 23. This response structure _____: (choose one)

- A. Speeds Excessively is a direct cause of Lose License
- B. Speeds Excessively is not a direct cause of Lose License
- C. Speeds Excessively is an indirect cause of Lose License
- D. Speeds Excessively is not an indirect cause of Lose License

 24. Consider the set of variables {Parents' Education, Intelligence, Education, Income}. Suppose you want to know whether _____ is a cause of Income, either direct or indirect. To answer this question, you need to examine the response structure that:

- A. Contains the variables Intelligence, Education, and Income, where Income is treated as the effect.
- B. Contains only the variables Education and Income, where Income is treated as the effect.
- C. Contains only the variables Intelligence and Income, where Income is treated as the effect.
- D. Contains only the variables Education and Income, where Education is treated as the effect.